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5.1.1. Stacked Plot

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Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

```
import matplotlib.pyplot as plt
```

```
import pandas as pd
```

```
# Data for Months and Temperature for three cities
```

```
data = {  
    'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August',  
             'September', 'October', 'November', 'December'],  
    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],  
    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],  
    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]  
}
```

```
# Write your code...
```

```
df = pd.DataFrame(data)

plt.stackplot(df['Month'],df['City_A_Temperature'],df['City_B_Temperature'],df['City_C_Temperature'])

plt.xlabel('Month')

plt.ylabel('Temperature')

plt.title('Temperature Variation')

plt.show()
```

